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Power charging system and control system for towing vehicle and towed vehicle connectable to towing vehicle

Abstract

A power charging system for a towing vehicle and towed vehicle combination for transferring high charging power from a charging device in the towing vehicle to an energy storage device on the towed vehicle. The power charging system is capable of safely transferring at least 1 kW of charging power. The power charging system includes a controller and sensors operably coupled to the controller via a communication system for safely powering the system when needed to charge the energy storage device, and safely de-powering the system prior to disconnecting the high-power charging circuit between the towing vehicle and the energy storage device.

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Background/Summary

BACKGROUND

(1) The field of the invention generally relates to power charging systems for charging an energy storage device, and more particularly, to power charging systems and methods for using a charging device in a towing vehicle to charge an energy storage device, such as a re-chargeable battery, in a towed vehicle connectable to the towing vehicle.

(2) There are a number of examples of towed vehicles which are disconnectably coupled to a

towing vehicle for supplying the primary motive power to transport the trailer. For example, in the trucking industry, tractor-trailer combinations are commonly used to transport cargo. The tractor (i.e., the “towing vehicle”) part of the combination includes a motor for supplying the motive power, and the trailer (i.e., the “towed vehicle”) is mechanically coupled to the tractor Similar towing-towed vehicle combinations include passenger vehicles (“towing vehicles”), such as automobiles and trucks, couplable to trailers, such as recreational vehicles (e.g., camper trailers, fifth wheelers, and the like) and moving trailers (“towed vehicles”).

(3) In each of these examples, the trailer may have a battery for powering various electrical devices and systems on the trailer, such as appliances like refrigerators, air conditioners, etc., when the towing vehicle's motor is not running or when the trailer is disconnected from the towing vehicle. Accordingly, the trailer may also be electrically coupled to the tractor using wires and/or cables to supply electrical power to subsystems and signals (e.g., turn signals, brake lights, etc.) on the trailer, and also to charge the trailer battery. Currently available power systems for such towing-towed vehicle combinations operate at 12 volts (12 V), which is a common standard for motor vehicles, or in more limited cases 24 V. For example, the power pins of towed vehicle common electrical plugs are able to carry at most 5-10 amps (5-10 A) of current. Accordingly, the electric power capability of the cabling, connectors and other circuitry are limited to relatively lower power levels in the range of 200 watts (200 W) to 400 watts (400 W) maximum. Moreover, current power systems for towing-towed vehicle combinations do not have the capability to fully sense the battery status, battery charging limitations or battery fault conditions. For example, many batteries, such as lithium-ion batteries (Li-Ion), have temperature limitations for charging and discharging, and can be damaged or even dangerous, if operated outside the correct parameters. Current power system typically only detect the battery voltage in controlling the battery charging status of the system.

(4) In addition, with the advent of high energy batteries, such as large Li-Ion battery modules, there is a need for supplying much higher power from the towed vehicle to the towing vehicle to charge the batteries. Current power systems are not capable of safely transmitting high power (e.g., above 1 kW) from between the towing vehicle and the towed vehicle. For example, the sockets on the connectors (e.g., plugs) on current power systems for electrically connecting the towed vehicle to the towing vehicle are currently used at relatively low voltage and low power levels, so they are not secured. If operated at higher power levels, these connectors would present a safety and performance risk of exposed and live voltage/current pins when not connected.

SUMMARY

(5) The presently disclosed inventions are directed to power systems for a towing vehicle and towed vehicle combination for transferring relatively high charging power from the towing vehicle to the towed vehicle to charge an energy storage device on the towed vehicle, which overcome the deficiencies of previously disclosed power systems, as described above. The towing vehicle may be any suitable vehicle including tractors, trucks, or automobiles, having a motive power source and a charging device. The towed vehicle may be any suitable trailer, including moving trailers, semi-trailers, truck trailers, camper trailers, fifth wheelers, and the like, and having an energy storage device, such as a chemical battery, Li-ion battery, capacitive storage device, solid state battery, or the like.

(6) In one embodiment, a power charging system according to the present invention includes a charging device installed on the towing vehicle. For example, the charging device may be an alternator on the towing vehicle which is a generator driven by the engine of the towing vehicle, or a battery on the towing vehicle, or other device that generates electric power. A towing vehicle interface is disposed on towing vehicle and is in selectable electrical connection with the charging device. For example, the power charging system may include a towing vehicle disconnect, such as a switch or relay, for selectably connecting and disconnecting the electrical connection between the charging device and the towed vehicle interface. The towed vehicle interface may be any suitable electrical connector, plug, or the like.

(7) The power charging system also includes an energy storage device installed on the towed vehicle. For example, the energy storage device may be a chemical battery, such as a lead acid battery, Li-ion battery, capacitive storage device, etc. The energy storage device is typically for powering devices like appliances, lighting, heating, ventilation and air conditioning (HVAC), and other electrical devices on the trailer. A towed vehicle interface is installed on the towed vehicle and is in selectable electrical connection with the energy storage device. For instance, a towed vehicle disconnect device, such as a switch or relay, may be used for connecting and disconnecting the electrical connection between the charging device and the towed vehicle interface. The towed vehicle interface is configured to be connected and disconnected with the towed vehicle interface and may be any suitable electrical connector, plug, or the like. The towed vehicle interface is not necessarily directly connected to the towing vehicle interface, but may be connectable to an adapter or cable connected to both the towed vehicle interface and the towing vehicle interface.
(8) The power charging system also includes a control system comprising a communication system and a controller. The communication system is configured to communicate operational data representative of operational conditions of the charging device and the energy storage device to a controller. The controller is operably coupled to the charging device and is configured to control the operation of the charging device based upon the operational characteristics. For instance, the controller may receive an energy storage device charge status indicating a charge level via the communication system, and the controller may then control whether the charging device is used to charge the energy storage device based on the charge status.
(9) The power charging system is configured to transfer charging power of least 1 kW from the charging device to the energy storage device. In other words, the towed vehicle interface and the towing vehicle interface, charging device, and any connectors, cables, pins, circuitry, etc. of the power charging system are properly sized and configured to safely transfer at least 1 kW of charging power from the charging device to the energy storage device. Alternatively, in additional aspects, the power charging system may be configured to transfer charging power from the charging device to the energy storage device of least 2 kW, or at least 4 kW, or at least 5 kW, or at least 7 kW. The higher transfer charging power allows faster charging and timely charging of ever larger energy storage devices. For instance, the energy storage device may have a capacity of at least 2 kW-hr, or at least 3.5 kW-hr, or at least 7 kW-hr or from 2 kW-hr to 50 kW-hr, or at least 50 kW-hr, or at least 75 kW-hr.
(10) In another aspect of the power charging system, the charging device comprises an alternator and an alternator controller. For instance, in one aspect of the invention, the alternator may be the chassis alternator of the towing vehicle. Alternatively, in another aspect, the alternator may be separate alternator to the chassis alternator which is dedicated to the power charging system and does not power or charge any towing vehicle equipment. The alternator controller may be a regulator which controls the operation of the alternator (e.g., controlling the field on the alternator).
(11) In still another aspect of the disclosed invention, the charging device may comprise an electrical power source and a power conversion device. For instance, in additional aspects, the electrical power source may be a battery, a lithium-ion battery, a lead acid battery, a capacitive storage device, a solid-state battery, an electrical vehicle battery, an alternator, a fuel cell, or any combination thereof. In yet another aspect, the power conversion device may be a DC-DC converter, a DC-AC converter, an AC-DC converter, a pulse width modulation controller, a current limiting wire, or a current limiting self-resetting device.
(12) In additional aspects, the towed vehicle interface and the towing vehicle interface may each comprise a single housing carrying both the charging power and operational data. For instance, each interface may be integrated into a single polymer housing which has high-power pins or contacts for carrying the charging power and auxiliary pins or contacts for carrying operational data. Alternatively, in another aspect, the towed vehicle interface and the towing vehicle interface may each comprise a power housing for carrying the charging power, and a physically separate data

housing for carrying at least some of the operational data and not carrying any of the charging power. As an example, each power housing may be a polymer housing having high-power pins or contacts for carrying the charging power, and each separate data housing may be a polymer housing having pins or contacts for carrying operational data but not charging power. In another aspect, at least some of the other operational data may not be carried by the data housing, but may be carried by yet other connector housings or even in simple signal wires, or wirelessly.

(13) In still another aspect, the communication system may be configured to communicate the operational data using a digital communication protocol. For example, the communication system may utilize a controller area network (CAN) having a CAN bus, or an Ethernet network, or other suitable digital communication protocol. In still another aspect, the communication system may comprise a wireless communication protocol, such as WiFi, Bluetooth, wireless USB, Zigbee, and cellular phone protocol.

(14) In another aspect of the power charging system, the signal wires may include analog sensing wires. For instance, the analog sensing wires may include one or more of a battery voltage sensing wire, a battery temperature wire, a thermocouple wire, a battery current sensing wire, a battery charge requested signal wire, and/or an encoded battery state signal wire.

(15) Alternatively, the signal wires may include one or more of a digital signal wire; a signal level wire, and an analog sensing wire, or any combination thereof. The signal wires carry operational data used by the controller to control the operation of the power charging system. Hence, in one aspect, the signal wires may include one or more of: a connection present signal wire which indicates whether the towing vehicle interface and towed vehicle interface are connected; a battery charging status signal wire which indicates whether charging of the battery is enabled, charging of the battery is disabled, and charge state to be maintained in which the battery the charging device supplies power to other loads in the trailer and only provides power to the battery to maintain a current state of charge.

(16) In another aspect, the operational data may include a charging status of the energy storage device, including a charging enable status and a charging disabled status. Accordingly, the charging status of the energy storage device indicates when the energy storage device is enabled to be charged, and when it is disabled from being charged. For instance, when the battery is not fully charged the charging status may indicate a charging enabled status, and when the battery is fully charged the charging status may indicate a charging disabled status. In another aspect, the charging status may also further include the charging needs of the energy storage device, such as how much energy is needed to fully charge the energy storage device, and/or a charging rate at which to charge the energy storage device.

(17) In still another aspect of the disclosed power charging system, the energy storage device may include a storage device management system (SDMS). In such case, the charging status may be determined by the SDMS and communicated from the SDMS to the controller. For instance, in the case of a battery as the energy storage device, the SDMS may be a battery management system (BMS). The SDMS may be a microcontroller having a processor, or it may be as simple as a logic circuit which provides a charge enable and charge disable signal to the controller to signal the controller whether to provide charging power to the energy storage device. Alternatively, the SDMS may be configured to communicate operational data regarding the energy storage device to the controller, and the charging status may be determined by the controller based upon the operational data obtained from the SDMS.

(18) In another aspect, the operational data may include present properties of the energy storage device, including present state of temperature, state of charge, capacity, voltage, and amperage into or out of energy storage device. In still other aspects, the operational data may further include specifications of the energy storage device, present needs of the charging device and energy storage device, and warnings, alarms and faults of both the charging device and energy storage device. In yet additional aspects, the operational data may also include the operational status of the charging

device, the specifications of the charging device, the operational status of the energy storage device, and the specifications of the energy storage device, where the operational status of the charging device includes one or more of the following properties of the charging device: current output; voltage output; temperature; overheating status; percentage utilization; operating mode, including standby, active, initializing, faulted state; identification of faults, including overheated, internal fault condition-logic error, battery not communicating, lost communication, broken sensor, battery fault; target voltage, energy storage voltage, target energy storage current; the operational status of the energy storage device includes one or more of the following properties of the energy storage device: present state of charge, present energy storage capacity; present voltage; present temperature; present current; charging mode, including charging enabled or charging disable; target energy storage voltage; target energy storage current; number of battery management systems (BMS's) present; whether energy storage device is online; fault state; and state of health; the specifications of the energy storage device include one or more of the following: type of energy storage device; overall storage capacity of battery; charging current limits of energy storage device; charging current limits of the BMS; charging voltage limits; charging temperature limits; and charging rate limits; and the specifications of the charging device include one or more of the following: output amperage capacity; output voltage; allowable operating temperature range; and power output capacity.

(19) In yet another aspect of the power charging device, the controller may be configured to selectably de-power the towing vehicle interface and the towed vehicle interface prior to disconnecting the towing vehicle interface from the towed vehicle interface. This feature can protect the charging device by avoiding rapid removal of the load on the charging device which may damage the charging device (e.g., damage to an alternator).

(20) In still another aspect, the controller may be configured to only power the towing vehicle interface and the towed vehicle interface when the towing vehicle interface is connected to the towed vehicle interface. For example, the controller can be configured to control switches between the towing vehicle interface and the charging device and between the towed vehicle interface and the energy storage device based on the connection status, and to only close the switches to power the respective interfaces when the controller detects that that the towing vehicle interface is connected to the towed vehicle interface. For instance, the power charging system may have a connection sensing device which detects when the towing vehicle interface and towed vehicle interface are connected, and the connection sensing device is operably coupled to the controller so the controller can determine whether there is a connection.

(21) In yet another aspect, the controller may be configured to selectably de-power the towing vehicle interface and the towed vehicle interface based upon a manually actuatable input by an operator. For example, the power charging system can have a button, switch, or other input device which an operator can actuate/select which signals the controller to de-power the towing vehicle interface and the towed vehicle interface (e.g., using switches, as described above).

(22) In additional aspects, the power charging system may also include a safe state indicator which indicates when it is safe to disconnect the towing vehicle interface and the towed vehicle interface upon de-powering of the towing vehicle interface and the towed vehicle interface. As some examples, the safe state indicator may be a light on one of towing vehicle interface and/or the towed vehicle interface, a light on the dashboard of the towed vehicle, a notification on a display device, or a notification on a software application (“app”) on a handheld computing device.

(23) In yet another aspect, the controller may be configured to selectably de-power the towing vehicle interface and the towed vehicle interface based upon an operational characteristic of the towing vehicle. For instance, the operational characteristic may be the towing vehicle being in park; the towing vehicle being in neutral, the towing vehicle being at idle, the towing vehicle parking brake being enabled, the towing vehicle engine being turned off, and/or the towing vehicle being below a maximum speed. Each of these operational characteristics may be predictive that the

power charging system is not going to be in further use, or that an operator may be disconnecting, or may wish to disconnect in the near future, the towing vehicle interface from the towed vehicle interface. Hence, de-powering the interfaces upon these operating events is a safety measure which prevents an operating from handling a powered interface.

(24) In additional aspects, the operational characteristics for de-powering the interfaces may be determined by one or more of: accessing the towing vehicle internal digital communications network; use of switches indicating position of transmission of the towing vehicle; use of switches indicating position of parking break mechanism of the towing vehicle; use of sensors to sense motion of the towing vehicle; use of sensors coupled to the charging device to indicate engine RPMs (revolutions per minute) of the towing vehicle; and use of sensors to indicate RPM of engine or transmission of the towing vehicle.

(25) In yet another aspect, the power charging system may also have a physical safety interlock on at least one of the towing vehicle interface and the towed vehicle interface. The physical safety interlock prevents disconnection of the towing vehicle interface and the towed vehicle interface until the power charging system, including the interfaces, charging device and/or energy storage device, are in a safe state for disconnection. In another aspect, the physical safety interlock may be operably coupled to, and controlled by the controller. In this way, the controller can detect whether the interfaces, charging device and/or energy storage device are in a safe state prior and only actuate the physical safety interlock to allow disconnection when they are in the safe state.

(26) In yet another aspect, the power charging system may also include a disconnect sensor configured to detect that the towing vehicle interface and the towed vehicle are being disconnected. For instance, the disconnect sensor may be disposed on at least one of the towing vehicle interface and the towed vehicle interface. The disconnect sensor may be operably coupled to the controller, and the controller can be configured to de-power the towing vehicle interface and the towed vehicle interface upon detecting that the towing vehicle interface and the towed vehicle are being disconnected.

(27) In still another aspect, the towing vehicle interface, the towed vehicle interface and the disconnect sensor may be configured such that the disconnect sensor detects that towing vehicle interface and the towed vehicle interface are being disconnected prior to a power connection between the towing vehicle interface and the towed vehicle interface carrying charging power from the charging device is disconnected. This allows the controller to de-power the towing vehicle interface and the towed vehicle interface prior to the power connection being disconnected, thereby preventing an unsafe condition in which one or both of the interfaces are powered and disconnected. For instance, the controller can be configured to de-power the towing vehicle interface and the towed vehicle interface based upon the disconnect sensor detecting that the towing vehicle interface and the towed vehicle are being disconnected prior to the power connection being disconnected.

(28) In another feature, the power charging device may also have a human user interface configured to communicate a status of the power charging system. As some examples, the human user interface may be a dashboard display on the towing vehicle; one or more indicator lights; a graphical display device; an LCD display; an LED display; an OLED display; and a wireless communication module configured to communicate with a software app on a handheld computing device. In yet another feature, the human user interface may be integrated with a fleet vehicle monitoring system to allow the fleet vehicle monitoring system to monitor the status of a fleet of towing vehicles and respective towed vehicles, including the status of the respective power charging systems. In one embodiment, the human user interface may be dedicated to the power charging system, or alternatively, it can be a human user interface of the towing vehicle or towed vehicle which is operably coupled to the power charging system.

(29) The power charging system may include any one or more of the aspects and features described herein, and need not include all of the various aspects and features. Accordingly, an improved

power charging system for a towing vehicle and a towed vehicle connectable to the towing vehicle is disclosed. The power charging system is capable of safely transferring high power levels, above 1 kW to 7 kW, from a charging device on the towing vehicle to an energy storage device on the towed vehicle. The power charging system may also be configured to safely de-power components of the system to prevent hazardous conditions which can create a risk of accidental electrical shock and/or damage to the system components.

(30) Another embodiment of the presently disclosed invention is directed to a subsystem, also referred to as a control system, for the power charging system disclosed herein. The subsystem may be an assembled device or module, or it may be a kit comprising a plurality of devices and/or module which can be assembled, connected, and/or installed into a power charging system. For instance, the subsystem may be installed into a towing vehicle and/or towed vehicle which already have a charging device and/or energy storage device installed.

(31) Accordingly, in one embodiment, a subsystem is disclosed for a power charging system comprising a charging device disposed on the towing vehicle and an energy storage device disposed on a towed vehicle connectable to the towing vehicle. For example, the power charging system may comprise any embodiment of the power charging system disclosed herein. The subsystem comprises a controller configured to be operable coupled to a communication system of the towing vehicle for receiving operational data representative of operational conditions of the charging device and the energy storage device. The controller is configured to operably couple to the charging device and to control the operation of the charging device based upon the operational data. The controller has an input connection for being connected to a storage device management system for the energy storage device. For instance, the input connection may be a CAN bus port. The controller is configured to detect whether the controller is connected to the storage device management system in order to determine whether a towing vehicle interface is connected to a towed vehicle interface. The controller is programmed (e.g., software and/or firmware) to only power the towing vehicle interface with the charging device when detecting a connection to the storage management device and to de-power the towing vehicle interface when detecting a disconnection of the controller from the storage device management system.

(32) In another aspect of the subsystem, the controller powers the towing vehicle interface by at least one of (i) powering on the charging device and (ii) connecting an output of the charging device to the towing vehicle interface, and de-powers the towing vehicle interface by at least one of (i) de-powering the charging device and (ii) disconnecting the output of the charging device to the towing vehicle interface.

(33) In another aspect, the charging device comprises a towing vehicle disconnect device for connecting and disconnecting the output of the charging device to the towing vehicle interface. In still another aspect, the towing vehicle disconnect device is one of a relay and a switch.

(34) In yet another aspect of the subsystem, the charging device comprises an alternator, and the controller is configured to function as a regulator for the alternator. The controller is configured powers on and de-power the charging device by regulating the alternator. In another aspect, the subsystem may further comprise the alternator.

(35) In still another aspect of the subsystem, the charging device may comprise an alternator and a regulator. The controller powers on and de-powers the charging device by controlling the regulator which in turn controls the charging device.

(36) In another aspect, the alternator may be a secondary alternator to a chassis alternator of the towing vehicle. The secondary alternator may be dedicated to the power charging system and is not configured for powering or charging any towing vehicle equipment.

(37) In another aspect, the charging device may comprise an alternator and a converter and the controller powers the towing vehicle interface by at least one of (i) powering on the converter, (ii) connecting an output of the converter to the towing vehicle interface, and de-powers the towing vehicle interface by at least one of (i) powering off the converter and (ii) disconnecting the output

of the converter from the towing vehicle interface. In still another aspect, the subsystem may further comprise the alternator and the converter.
(38) In another aspect, the connection that the controller is configured to detect to determine whether the controller is connected to the storage device management system is one of a CAN bus connection, an Ethernet connection and an RS-485 connection.
(39) In still another aspect, the subsystem further comprises a towed vehicle disconnect device electrically connected between the energy storage device and a towed vehicle interface which is configured to be connected to the towing vehicle interface. The towed vehicle disconnect device is configured to electrically connect and disconnect the electrical connection between the energy storage device and the towed vehicle interface.
(40) In another aspect, the towing vehicle interface may include a disconnect device connection connected to a towed vehicle disconnect circuit. The towed vehicle disconnect device is configured to connect a connection between the energy storage device and the towed vehicle interface when the towed vehicle interface is connected to the towed vehicle disconnect circuit, and to disconnect the connection between the energy storage device and the towed vehicle interface when the towed vehicle interface is disconnected from the disconnected device circuit. In still another aspect, the towed vehicle disconnect device may comprise one of a switch and a relay, and the towed vehicle disconnect circuit is connected to one of a ground wire or a hot wire on the towing vehicle and the towed vehicle disconnect device is connected to the other of a ground wire or a hot wire on the towed vehicle.
(41) In another aspect of the subsystem, the controller is configured to selectably de-power the towing vehicle interface based upon an operational characteristic of the towing vehicle. For example, the operational characteristic may be one or more of: the towing vehicle being in park; the towing vehicle being in neutral; the towing vehicle being at idle; the towing vehicle parking break being enabled; the towing vehicle engine being turned off; and the towing vehicle being below a maximum speed.
(42) In another aspect, the operational characteristic may be determined by one or more of: accessing a towing vehicle internal digital communications network; use of switches indicating position of transmission of the towing vehicle; use of switches indicating position of parking break mechanism of the towing vehicle; use of sensors to sense motion of the towing vehicle; use of sensors coupled to the charging device to indicate engine RPMs of the towing vehicle; and use of sensors to indicate RPM of engine or transmission of the towing vehicle.
(43) In another aspect, the controller de-powers the towing vehicle interface by at least one of (i) powering on the charging device and (ii) connecting an output of the charging device to the towing vehicle interface, and de-powers the towing vehicle interface by at least one of (i) de-powering the charging device and (ii) disconnecting the output of the charging device to the towing vehicle interface.
(44) In yet another aspect, the charging device comprises a towing vehicle disconnect device for connecting and disconnecting the output of the charging device to the towing vehicle interface. In another feature, the towing vehicle disconnect device is one of a relay and a switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing and other aspects of embodiments are described in further detail with reference to the accompanying drawings, wherein like reference numerals refer to like elements and the description for like elements shall be applicable for all described embodiments wherever relevant, wherein:
(2) FIG. **1** is a block diagram of a power charging system for a towing vehicle and towed vehicle

combination for transferring high charging power, according to one embodiment of the present invention;
(3) FIG. **2** is a side view of a schematic of a single housing embodiment of the towing vehicle interface and towed vehicle interface which may be utilized in the power charging system of FIG. **1**;
(4) FIG. **3** is a side view of a schematic of a dual housing embodiment of the towing vehicle interface and towed vehicle interface which may be utilized in the power charging system of FIG. **1**;
(5) FIG. **4** is an electrical schematic of the power charging system of FIG. **1**, according to one embodiment

DETAILED DESCRIPTION

(6) Referring to FIG. **1**, a block diagram of one exemplary embodiment of a power charging system **100** for a towing vehicle **102** and towed vehicle **104** combination for transferring high charging power from the towing vehicle **102** to the towed vehicle **104** is illustrated. The power charging system **100** is shown as installed in a towing/towed vehicle combination **106**. The towing vehicle **102** may be any type of vehicle, including for example, tractors, trucks, automobiles, motor vehicles, etc., having a motor (engine) which may be an internal combustion engine, electric motor, or other suitable motor. The towed vehicle may be any type of trailer, including for example, cargo trailers, camper trailers, utility trailers, livestock trailers, boat trailers, equipment trailers, fifth wheelers, car haulers, etc. The towing vehicle **102** has a trailer hitch **108** which couples to a trailer coupler **110** on the towed vehicle **104** for providing the mechanical connection between the towing vehicle **102** and the towed vehicle **104**. The trailer hitch **108** and trailer coupler **110** may be any suitable hitch/coupler combination, such as a ball/receiver, fifth-wheel/king-pin, pintle/ring, etc.
(7) The power charging system **100** includes a charging device **112** installed on the towing vehicle **102**. The charging device **112** provides the charging power for the system **100**, which is typically electric power. The charging device **112** may be an alternator on the towing vehicle **102**. An alternator is an electric generator which is driven by the motor of the towing vehicle to produce electric power. The charging device **112** may also be battery (such as an accessory battery disposed on the towing vehicle **102**), a lithium-ion battery, a lead acid battery, a capacitive storage device, a solid-state battery, an electrical vehicle battery, a fuel cell, or any combination of the charging devices **112** disclosed herein.
(8) The charging device **112** may also include a converter **114** which converts the output electric power from the charging device **112**, such as the output electric power from an alternator or a battery, to a different voltage or from direct current (DC) to alternating current (AC) or vice versa. For example, the charging device **112** may be an alternator which outputs AC electric power at an output voltage, and the converter converts the AC electric power to DC electric power at a converted voltage, typically about 14 V in standard automobiles. In the case of the charging device **112** comprising an alternator, the charging device **112** may also include a regulator, also called an alternator controller, which controls the operation of the alternator (e.g., controlling the field on the alternator). Hence, the converter **114** may be a DC-DC converter, a DC-AC converter, an AC-DC converter, a pulse width modulation controller, a current limiting wire, and a current limiting self-resetting device.
(9) The power charging system **100** may also include an optional buffering device **111**. The buffering device **111** is configured to store charging power from the charging device **112**, and to provide the charging power to the power charging system **100**, for example, when the charging device **112** is unavailable to provide charging power. Hence, the buffering device **111** may be any suitable battery, such as lead acid battery, Li-ion battery, a capacitive storage device, a solid-state battery, or other storage device configured to store power and output the stored power as electric power. Thus, in the case of power charging system **100** having a buffering device **111**, wherever it is described that the charging device **112** provides electric power, such electric power may be

provided by the buffering device **111**. For instance, when the charging device **112** is unavailable (e.g., when an alternator is not producing electric power because the towing vehicle motor is not running).

(10) In the case of a charging device **112** comprising an alternator, the alternator can be the chassis alternator of the towing vehicle **102**. In other words, the alternator on the towing vehicle **102** which provides electrical power to the towing vehicle **102** for powering the electrical systems and components of the towing vehicle **102**, including charging a battery on the towing vehicle **102**. Alternatively, the alternator may be a dedicated alternator dedicated to the power charging system **100** which is used only to provide electric power to the power charging system **100** and which does not power any other equipment of the towing vehicle **102**. The dedicated alternator is also driven by the motor of the towing vehicle **102**. The dedicated alternator may also have a separate regulator for controlling the operation of the dedicated alternator (e.g., controlling the field on the dedicated alternator). Alternatively, a single regulator may be configured to control both the chassis alternator and the dedicated alternator.

(11) The power charging system **100** also includes a towing vehicle disconnect device **118** electrically connected between the charging device **112** and a towing vehicle interface **120**. The towing vehicle disconnect device **118** is configured to selectably electrically connect and disconnect the electrical connection between the charging device **112** and the towing vehicle interface **120**. The towing vehicle disconnect device **118** may be a switch or relay, or the like.

(12) The towing vehicle interface **120** is a connector for electrically connecting the charging device **112** on the towing vehicle **102** to an energy storage device **122** on the towed vehicle **104**. The towing vehicle interface **120** is configured to be connected and disconnected with a towed vehicle interface **124** on the towed vehicle **104**. The towing vehicle interface **120** may be any suitable electrical connector, plug, or the like. The towing vehicle interface **120** and towed vehicle interface **124** may not directly connect to each other, but may each be connectable to opposite ends of a connection cable **126** or an adapter for completing the electrical connection between the towing vehicle interface **120** and the towed vehicle interface **124**. The connection cable **126** has a towing vehicle connector **128** on a first end of the connection cable **126** and connectable to the towing vehicle interface **120**, and a towed vehicle connector **130** on a second end of the connection cable **126**.

(13) The power charging system **100** also has a controller **132** for controlling the operation of the power charging system **100**. The controller **132** is a computing system comprising a microprocessor, memory, input and output connections and buses, one or more communication adapters, one or more storage devices (e.g., hard drive, memory, etc.), and software and firmware which programs the controller **132** to perform the functions as described herein. The controller **132** is operably coupled to the charging device **112**, the towing vehicle disconnect device **118**, and may also be operably coupled to various other components of the power charging system **100**, as described herein. The controller **132** is configured to control the operation of the charging device **112** based upon operational data received by the controller **132**, such as operational data from various sensors and/or management systems on the towing vehicle **102** and/or towed vehicle **104**. For example, the controller **132** is operably coupled to towing vehicle sensors **136**, towed vehicle sensors **137**, a vehicle computer **136**, and/or a storage device management system (SDMS) **138** via a communication system **140**. As described in more detail herein, each of these components can provide operational data to the controller **132** which the controller **132** can use to control the operation of the charging system **100**, including the operation of the charging device **112**, the towing vehicle disconnect device **118** and/or a towed vehicle disconnect device **134**. In the case that the charging device is an alternator, the controller **132** may be integrated with, or operably coupled to, the regulator to control the operation of the alternator, including enabling and disabling the alternator, and controlling the power output of the alternator, based upon the operational data received by the controller **132**. As described in more detail below, the controller **132** also controls

the operation of the towing vehicle disconnect device **118** and/or the towed vehicle disconnect device **134**.

(14) The power charging system **100** also has a communication system **140** for providing communication between one or more of the components of the power charging system **100**, including between the controller **132**, towing vehicle sensors **136**, towing vehicle sensors **137**, a vehicle computer **136**, (SDMS) **138** and/or a sub-controller **133**. The communication system **140** may include a communication network operably coupled to each of these components for providing communication between the components. As an example, the communication network may comprise a CAN network in which each of the components is connected to the network via a two wire CAN bus **142**. Alternatively, the communication system **140** may comprise an Ethernet network, a RS-485, wireless communication network having a wireless communication protocol (such as WiFi, Bluetooth, wireless USB, Zigbee, cellular phone protocol, etc.). The communication system **140** may also include one or more signal wires **144** operably coupling the controller **132** to other components, such as to the towing vehicle sensors **134**, towed vehicle sensors **137**, and/or sub-controller **133**. The signal wires **144** may include analog sensing wires, including one or more of a battery voltage sensing wire (connected to a battery voltage sensor), a battery temperature wire (connected to a battery temperature sensor), a thermocouple sensing wire (connected to a thermocouple), a battery current sensing wire (connected to a current sensor), an energy storage device charge requested signal wire (connected to the SDMS **138** and/or sub-controller **133**), and an encoded battery state signal wire (connected to the SDMS **138** and/or sub-controller **133**). The signal wires **144** may also include CAN bus wires **142** (e.g., CAN twisted pair), analog level wires, digital signal wires, Ethernet cables, etc.

(15) Turning to the components of the power charging system **100** on the towed vehicle **102**, the energy storage device **122** is an electrically re-chargeable device for storing energy and outputting the stored energy as electric power for powering equipment on the towed vehicle **102**, such as appliances, lighting, heating, ventilation and air conditioning (HVAC), and other electrical devices on the trailer and/or auxiliary equipment outside the towed vehicle **102**, like camping equipment, pumps, lights, cooking equipment, etc. As some non-limiting examples, the energy storage device **122** may be a chemical battery, such as a lead acid battery, Li-ion battery, a capacitive storage device, a solid-state battery, etc.

(16) The towed vehicle interface **124** is configured to be connected and disconnected with the towed vehicle interface **124** on the towed vehicle **104**. The towed vehicle interface **124** may be any suitable electrical connector, plug, or the like. As depicted in the example of FIG. **1**, the towing vehicle interface **120** and towed vehicle interface **124** may not directly connect to each other, but may each be connectable to opposite ends of a connection cable **126** or an adapter for completing the electrical connection between the towing vehicle interface **120** and the towed vehicle interface **124**. Alternatively, the towed vehicle interface **124** and towing vehicle interface **120** may be configured to connect directly to each other.

(17) The towed vehicle disconnect device **134** is configured to selectably electrically connect and disconnect the electrical connection between the energy storage device **122** and the towed vehicle interface **124**. The towed vehicle disconnect device **134** may be a switch or relay, or the like.

(18) The power charging system **100** may also have a sub-controller **133** for controlling the operation of the power charging system **100**. The sub-controller **133** may be a part of the controller **132** that operates in conjunction with the controller **132**, or it may be a separate module which is in operable communication with controller **132**, or it may be integrated with the controller **132** as an integrated module. The sub-controller **133** is operably coupled to the charging device energy storage device **122**, the towed vehicle disconnect device **134**, and may also be operably coupled to various other components of the power charging system **100**, as described herein. The sub-controller **133** is configured to control the operation of the towed vehicle disconnect device **134** based upon signals and/or operational data received from the controller **132** and/or from various

sensors and/or management systems on the towing vehicle **102** and/or towed vehicle **104**. For example, the sub-controller **133** may be operably coupled to the SDMS **138** and/or the towed vehicle sensors **137** via the communication system **140**. As described in more detail herein, each of these components can provide operating signals and/or operational data to the sub-controller **133** which it can use to control the operation of the charging system **100**, including the operation of the energy storage device **122** and/or the towed vehicle disconnect device **134**. It should be understood that the controller **132** and sub-controller **133** may be installed on either the towing vehicle **102** or the towed vehicle **104**, as they can communicate with components on either towing vehicle **102** or the towed vehicle **104** via the communication system **140**.
(19) The power charging system **100** also has one or more operator interfaces **146** for providing outputs to an operator and/or for receiving inputs from an operator. The operator interfaces **146** include one or more safe state indicators **148** which indicate when it is safe to disconnect the towing vehicle interface **120**, the towed vehicle interface **124** and/or the connection cable **126** upon de-powering of the towing vehicle interface **120** and the towed vehicle interface **124**. The safe state indicator **148** may be any suitable indicator including, without limitation, a light **160** on the towing vehicle interface **120**, a light **162** on the towed vehicle interface **124**, a light on the dashboard of the towed vehicle, a notification on a display device **152**, and/or a notification on a software app on a portable computing device **154**. The portable computing device **154** may be any suitable computing device, including without limitation, a laptop computer, a tablet computer, a handheld computing device, a cellular phone, a smartphone, etc.
(20) The operator interfaces **146** may also include a human user interface **150** having a display device **152**. The human user interface **150** is configured to communicate a status of the power charging system **100** to an operator, and also allow the operator to make inputs to the power charging system **100** to control the operation of the power charging system **100**. The human user interface **150** may be a system dedicated to the power charging system **100**, or it may be integrated with a human interface system of the towing vehicle **102** or the towed vehicle **104**. For example, the human user interface **150** may be integrated with the towing vehicle computer **138** has a vehicle display and an input device such as a touchscreen, touchpad, joystick, or the like. The human user interface **150** may comprise a computing device having a graphical display device **152** such as an LCD display, an LED display or an OLED display, a dashboard display on the towing vehicle **102**, one or more indicator lights, and/or a wireless communication module **156** configured to communicate with the portable computing device **154** having software app for interfacing with the power charging system **100**.
(21) The power charging system **100** may also be in communication and integrated with a fleet vehicle monitoring system **157** to allow the status of the charging system **100** to be monitored along with a fleet of vehicles having installed power charging systems **100**. In one way, the human user interface **150** may be in communication with the fleet vehicle monitoring system via a wireless communication network **158**.
(22) Turning to FIGS. **2** and **3**, schematic examples of embodiments of the towing vehicle interface **120**, towed vehicle interface **124** and connection cable **126** are illustrated. FIG. **2** illustrates a single housing embodiment for carrying both the charging power and for carrying the signals/data through the towed vehicle interface **120**, towing vehicle interface **124** and connection cable **126**. In the embodiment of FIG. **2**, each of the towing vehicle interface **120**, the towed vehicle interface **124** and the towing vehicle connector **128** and towed vehicle connector **130** of the connection cable **126** comprises a single housing for carrying the charging power in the power wires **164** and for carrying the signals/data in the signal/data wires **166**. Each of the towing vehicle interface **120**, the towed vehicle interface **124**, the towing vehicle connector **128** and towed vehicle connector **130** have high-power pins or contacts **172** for carrying the charging power and auxiliary pins **174** for carrying the operational data and signals from the signal wires **166**.
(23) The towing vehicle interface **120** may also have a towing vehicle disconnect sensor **176** which

detects whether the towing vehicle interface **120** is connected or disconnected, and/or is being disconnected, from the towed vehicle interface **124** and/or the towing vehicle connector **128**. The towing vehicle disconnect sensor **176** may be configured such that it can detect when the towing vehicle interface **120** is being disconnected, i.e., is in the process of being disconnected, prior to the high-power pins/contacts **172** of the towing vehicle interface **120** disconnecting from the mating high-power pins/contacts of the towing vehicle connector **128**.

(24) Similarly, the towed vehicle interface **124** may have a towed vehicle disconnect sensor **178** which detects whether the towing vehicle interface **120** is connected or disconnected, and/or is being disconnected, from the towed vehicle interface **124** and/or the towed vehicle connector **130**. The towed vehicle disconnect sensor **178** is same or similar, and may have the same functionality, as the towing vehicle disconnect sensor **176**. In one embodiment, the towing vehicle disconnect sensor **178** may simply be a conductive wire which connects to a ground when the towing vehicle interface **120** is connected to the towed vehicle interface **124**, for example, see the signal wire **144***v* depicted in the electrical schematic of FIG. **4**, and described below.

(25) FIG. **3** illustrates a dual housing embodiment for carrying the charging power and the signals/data between the towing vehicle **102** and the towed vehicle **104**. There is a first housing for each of a first towing vehicle interface **120***a*, a first towed vehicle interface **124***a*, and a first towing vehicle connector **128***a* and a first towed vehicle connector **130***a* of a first connection cable **126***a* for carrying the charging power. Accordingly, each of the first towing vehicle interface **120***a*, the first towed vehicle interface **124***a*, the first towing vehicle connector **128***a* and first towed vehicle connector **130***a* have high-power pins or contacts **172** for carrying the charging power. There is a physically separate second housing for each of a second towing vehicle interface **120***b*, a second towed vehicle interface **124***b*, and a second towing vehicle connector **128***b* and second towed vehicle connector **130***b* of a second connection cable **126***b* for carrying signals/data in the signal/data wires **166**. Each of the second towing vehicle interface **120***b*, the second towed vehicle interface **124***b*, the second towing vehicle connector **128***b* and the second towed vehicle connector **130***b* have auxiliary pins or contacts **172** for carrying signals/data in the signal/data wires **166**. The second housings of the second towing vehicle interface **120***b*, the second towed vehicle interface **124***b*, the second towing vehicle connector **128***b* and the second towed vehicle connector **130***b* do not carry charging power. However, the first housing of the first towing vehicle interface **120***a*, the first towed vehicle interface **124***a*, the first towing vehicle connector **128***a* and first towed vehicle connector **130***a* may also carry signals/data from other signal wires.

(26) The towing vehicle interface **120** also has a physical safety interlock **168** which prevents disconnection of the towing vehicle interface **120** from the towed vehicle interface **124** via the connection cable **126**, if utilized, until the power charging system **100** is in a safe state for disconnection. For instance, the physical safety interlock **168** may be actuated to prevent disconnection until the electric power is removed from the towing vehicle interface **120**, such as by opening the towing vehicle disconnect device **118** and the towed vehicle disconnect device **134** and/or disabling the charging device **112** (e.g., disabling the alternator to prevent damage to the alternator from removing the load on the alternator). The physical safety interlock **168** may be operably coupled to, and actuated by, the controller **132**, or it may be self-actuatable in response to detecting whether there is electric power at the towing vehicle interface **120**.

(27) The towed vehicle interface **124** also has a physical safety interlock **170** which prevents disconnection of the towed vehicle interface **124** from the towing vehicle interface **120** via the connection cable **126**, if utilized, until the power charging system **100** is in a safe state for disconnection. For instance, the physical safety interlock **170** may be actuated to prevent disconnection until the electric power is removed from the towed vehicle interface **124**, such as by opening the towing vehicle disconnect device **118** and the towed vehicle disconnect device **134** and/or disabling the charging device **112** (e.g., disabling the alternator to prevent damage to the alternator from removing the load on the alternator). The physical safety interlock **168** may be

operably coupled to, and actuated by, the controller **132**, or it may be self-actuatable in response to detecting whether there is electric power at the towing vehicle interface **120**.

(28) Referring now to FIG. **4**, an electrical schematic of the power charging system **100** is illustrated. The electrical schematic of FIG. **4** includes many of the possible electrical connections of components of the power charging system **100**, including analog and digital connections. The electrical schematic of FIG. **4** is not intended to illustrate an actual embodiment of the power charging system **100**, but is intended to show the various possible connections, only some of which would be used in actual power charging system. Because of the high number of possible enabling combinations of electrical connections, it is not possible to show an electrical schematic for each and every combination. Accordingly, some of electrical connections shown in FIG. **4** are redundant, and not all of the electrical connections are required for an operable power charging system **100**. With the descriptions provided herein, one of ordinary skill in the art will be enabled to determine which electrical connections are needed based on the components utilized in a power charging system **100**, to the full breadth of the inventions disclosed and claimed herein.

(29) The controller **132** is in operable communication with the components of the power charging system **100** via the communication system **140** which comprises a plurality of signal wires **144**, and a wireless communication module **156**. As described above, the signal wires **144** may include analog sensing wires, CAN bus wires **142**, analog level wires, digital signal wires, Ethernet cables, or other suitable communication wiring, etc. Unless specifically described otherwise, the signal wires **144** may be any of the types of wires described herein. Furthermore, any of the described signal wires **144** may be one or more wires, which are described as a single signal wire **144**.

(30) As illustrated in FIG. **4**, the charging device **112** is connected to the controller **132** via signal wire **144***a*. The charging device **112** is also connected to the regulator **116** via signal wire **144***b*. The regulator **116** is connected to the controller via signal wire **144***c* for controlling the operation of the charging device **112**. An engine rpm sensor **136***e* is operably coupled to the charging device **112** for detecting the engine rpm. The engine rpm sensor **136***e* is connected to the controller **132** via signal wire **144***d* for transmitting a signal from the engine rpm sensor **136***e* to the controller **132**. The towing vehicle computer **138** is connected to the controller **132** via signal wire **144***e*, which may be a CAN bus wire **142**, for transceiving operational data and signals between the towing vehicle computer **138** and the controller **132**.

(31) Each of the towing vehicle sensors **136** may be directly connected to the controller **132** to provide analog signals, analog level signals, or digital signals. The towing vehicle sensors **136** transmit a signal representative of certain operational data to the controller **132**, as described herein. The transmission position sensor **136***a* is connected to the controller **132** via signal wire **144***f*. The transmission position sensor **136***a* detects the position of the transmission of the towing vehicle **102**, such as whether the towing vehicle **102** is in drive, park, neutral, etc. The parking break position sensor **136***b* is connected to the controller **132** via signal wire **144***g*. The parking break position sensor **136***b* detects whether the parking break is actuated or de-actuated. The vehicle motion sensor/speedometer **136***c* is connected to the controller **132** via signal wire **144***h*. The vehicle motion sensor/speedometer **136***c* detects whether the towing vehicle **102** is moving and/or the speed of the towing vehicle **102**. The engine RPM sensor **136***d* is connected to the controller **132** via signal wire **144***v*. The engine RPM sensor **136***d* detects the RPMs of the engine.

(32) In addition, some or all of the towing vehicle sensors **136** may be connected to the towing vehicle computer **136** which receives the signals from the towing vehicle sensors **136**. The towing vehicle computer **136** can then transmit the signals from such towing vehicle sensors **136** to the controller via the signal wire **144***e*.

(33) Each of the operator interfaces **146**, including the safe state indicator **148**, portable computer **154**, human user interface **150** and display device **152** are connected to the controller **132** via the communication system **140**, such as additional signal wires **144**, a wireless communication module **156**, and/or other communication network.

(34) The towing vehicle disconnect device **118** is connected to the controller **132** via signal wire **144***i*. The controller **132** is configured to actuate and de-actuate the towing vehicle disconnect device **118** via the signal wire **144***i* to selectably electrically connect and disconnect the electrical connection between the charging device **112** and the towing vehicle interface **120** based on operational data received by the controller **132**. The towed vehicle disconnect device **134** is connected to the sub-controller **133** via signal wire **144***x*. The towed vehicle disconnect device **134** may also be connected to the controller **132** via signal wire **144***y*. The sub-controller **133** and/or controller **132** are configured to actuate the towed vehicle disconnect device **134** to selectably electrically connect and disconnect the electrical connection between the energy storage device **122** and the towed vehicle interface **124** based on operational data received by the controller **132** and/or sub-controller **133**. In an alternative embodiment, the towed vehicle disconnect device **134** may simply be connectable to a disconnect device connection of the towing vehicle interface **118** which is in turn connected to a towed vehicle disconnect circuit. The towed vehicle disconnect device **134** is then configured to connect a connection between the energy storage device **122** and the towed vehicle interface **124** when the towed vehicle interface **124** is connected to the towed vehicle disconnect circuit (i.e., when the towed vehicle interface **124** is connected to the towing vehicle interface **120**, such as by connecting the connection cable **126**), and to disconnect the connection between the energy storage device **122** and the towed vehicle interface **124** when the towed vehicle interface **124** is disconnected from the disconnect device circuit. For instance, the towed vehicle interface **124** may be relay in which one pole is connected to a hot wire or ground wire on the towed vehicle **104** and the other pole is selectably connected and disconnected to the towed vehicle disconnect circuit which is connected to the other of a ground wire or a hot wire on the towed vehicle **102**.
(35) A local voltage sensing signal wire **144***j* is connected to the power wires **164** and the controller **132** to allow the controller **132** to detect the voltage across the power wires **164** at a position between the towing vehicle disconnect **118** and the energy storage device **122**.
(36) The safe state indicator light **160** is connected to the controller **132** via signal wire **144***j* **148**. The safe state indicator light **162** is connected to the controller **132** via signal wire **144***k*. The SDMS is connected to the controller **132** via signal wire **144***m*. The sub-controller **133** is connected to the controller **132** via CAN bus wire **142**. The SDMS **138** may also be connected to the sub-controller **133** such that the SDMS **138** communicates to the controller **132** via the sub-controller **133**.
(37) Each of the towed vehicle sensors **137** may be directly connected to the controller **132** to provide analog signals, analog level signals, or digital signals. Each of the towed vehicle sensors **137** transmits a signal representative of certain operational data to the controller **132**, as described herein. The battery voltage sensor **137***a* is connected to the controller **132** via signal wire **144***p*. The battery voltage sensor **137***a* detects the voltage of the energy storage device **122**. The battery temperature sensor **137***b* is connected to the controller via signal wire **144***q*. The battery temperature sensor **137***b* detects the temperature of the energy storage device **122**. The thermocouple wire **137** is connected to the controller **132** via signal wire **144***r*. The thermocouple wire **137** detects a temperature, such as the temperature of the energy storage device **122**. The battery current sensor/shunt **137***d* is connected to the controller **132** via signal wire **144***s*. The battery current sensor/shunt **137***d* detects the current flowing at the output of the energy storage device **137***d*. The battery charge requested signal device **137***e* is connected to the controller **132** via signal wire **144***t*. The battery charge requested signal device **137***e* provides a signal to the controller **132** as to whether the energy storage device **122** is requesting to be charged, and may be a signal level wire in which the controller **132** determines whether the energy storage device **122** is requesting to be charged based on the voltage level, or a digital signal wire which provides a digital signal to the controller **132** as to whether the energy storage device **137** is requesting to be charged.
(38) The encoded battery state signal device **137***f* is connected to the controller **132** via signal wire

144*u*. The encoded battery state signal device **137***f* detects a state of the energy storage device **122**, including one or more of the following: overheating status; percentage utilization; operating mode, including standby, active, initializing, faulted state; identification of faults, including overheated, internal fault condition-logic error, battery not communicating, lost communication, broken sensor, battery fault; target voltage, energy storage voltage, target energy storage current; present state of charge, present energy storage capacity; present voltage; present temperature; present current; charging mode, including charging enabled or charging disable; target energy storage voltage; target energy storage current; number of SDMS's (e.g., BMS's) present; whether energy storage device is online; fault state; and state of health; type of energy storage device; overall storage capacity of battery; charging current limits of energy storage device; charging current limits of the BMS; charging voltage limits; charging temperature limits; and charging rate limits; and the specifications of the charging device include one or more of the following: output amperage capacity; output voltage; allowable operating temperature range; and power output capacity. The encoded battery state signal device **137***f* may also be a battery charging status signal wire which indicates whether charging of the battery is enabled, charging of the battery is disabled, and/or a charge state to be maintained in which the charging device supplies power to other loads in the trailer and only provides power to the battery to maintain a current state of charge.

(39) An electrical ground **180** on the towed vehicle **104** is connected to the controller **132** via signal wire **144***w*. The signal wire **144***w* and electrical ground **180** can be used by the controller **132** to determine whether there is a connection between the towing vehicle interface **120** and the towed vehicle interface **124**. For example, the controller **132** can determine whether the connection cable **126** is fully connected to the towing vehicle interface **120** and the towed vehicle interface **124**.

(40) Furthermore, some or all of the towed vehicle sensors **137** may be connected, and/or integrated with the SDMS **138**, instead of being connected to the controller **132**, or in addition to being connected to the controller **132**. The SDMS **138** receives the signals from the towed vehicle sensors **137**, and the SDMS can then transmit the signals from such towed vehicle sensors **137** to the controller **132** via the signal wire **144***m*. Also, the SDMS **138** may be connected to the sub-controller **133**, in which case, the SDMS **138** can transmit any signals from the SDMS and/or from the towed vehicle sensors **137**. Then, the sub-controller **133** can transmit such signals to the controller **132** via the CAN bus signal wires **142**.

(41) The power charging system **100** is configured to safely transfer high charging power from the charging device **112** to the energy storage device **122** for charging the energy storage device **122**. All of the components carrying the charging power are configured to safely handle the high charging power of the power charging system **100**, including without limitation, any one or more of the charging device **112**, the converter **114**, the power wires **164**, the towing vehicle disconnect device **118**, the towing vehicle interface **120**, the towing vehicle connector **128**, the connection cable **126**, the towed vehicle connector **130**, the towed vehicle interface **124**, the towed vehicle disconnect device **134**, the energy storage device **122**, the SDMS **138**, and/or any others connectors, cables, pins, circuitry, etc. of the power charging system **100** which carry the charging power. Accordingly, in one embodiment, the power charging system **100** is rated to transfer at least 1 kW of charging power from the charging device **112** to the energy storage device **122** for charging the energy storage device **122**. In alternative embodiments, the power charging system **100** is rated to transfer at least 1.5 kW, or at least 2 kW, or at least 5 kW, or at least 7 kW of charging power from the charging device **112** to the energy storage device **122** for charging the energy storage device **122**. In order to transfer at these high charging powers, the power charging system **100** may charge at higher voltages than the typical 12 V of most towing vehicle electric systems. Hence, in embodiments of the presently disclosed power charging system **100**, the power charging system **100** provides charging power from the charging device **112** and/or converter **114** at about 24 V, or at about 36 V, or at about 48 V, or at least about 20V, or at least about 24 V, or at least about 48V. The term “about” with respect to the charging voltage means within plus or minus

10% of the stated voltage. The higher transfer charging power allows faster and more timely charging of larger energy storage devices **122**. In various embodiments of the power charging system **100**, the energy storage device **122** may have an energy storage capacity of at least 2 kW-hr, or at least 3.5 kW-hr, or at least 7 kW-hr or from 2 kW-hr to 50 kW-hr, or at least 50 kW-hr, or at least 75 kW-hr.

(42) The operation and functionality of the power charging system **100** and its components will now be described. As described herein, the power charging system **100** can safely transfer high charging power from the charging device **112** on the towing vehicle **102** to the energy storage device on the towed vehicle **104**. In order to safely transfer the high charging power, the power charging system **100** is configured to safely handle high charging power, and also to monitor various conditions and characteristics of the power charging system **100** and control the operation of the power charging system **100** based on conditions and characteristics to avoid unsafe conditions. More specifically, the power charging system **100** uses the communication system **140** and operational data from the various towing vehicle sensors **136**, the towing vehicle computer **138**, the charging device **112**, the operator interfaces **146**, the towed vehicle sensors **137**, the SDMS **138**, the energy storage device **122**, and/or the sub-controller **133** to provide such operational data to the controller **132**, and the controller **132** is configured to control the operation of the power charging system **100** based on such operational data to avoid unsafe operating conditions. As some examples, the controller **132** may be configured to power the towing vehicle interface **120** and towed vehicle interface **124** only when they are connected to each other such that the power charging system **100** avoids unsecured, exposed and dangerous high voltage and high current connectors. The controller **132** may also be configured to control the charging power delivered to the energy storage device **122** based on the conditions of the energy storage device **122**, such as state of charge, specifications of the energy storage device (e.g., charging current limits, charging voltage limits, charging temperature constraints, etc.), faults conditions and warnings regarding the energy storage device **122**.

(43) Accordingly, in the starting state of the power charging system **100** with the towed vehicle **104** disconnected from the towing vehicle **102**, and the vehicle parked and the vehicle motor not running, the towing vehicle interface **120** and the towed vehicle interface **124** are not powered (i.e., there is no charging power at the interface **120** and **124**). This may be accomplished by the controller **132** detecting that the towing vehicle interface **120** is not connected to the towed vehicle interface **124** from signals from the signal wire **144***v* (i.e., the controller detecting that the controller **132** is not grounded via the signal wire **144***v*), the towing vehicle disconnect sensor **176**, and/or the towed vehicle disconnect sensor **178**. When detecting that the towing vehicle interface **120** is not connected to the towed vehicle interface **124**, the controller **132** maintains the disconnect device **118** (assuming that the disconnect device **118** is normally open (disconnected state) and closes (connected state)) in the disconnected state which disconnects the connection between the charging device **112** and the towing vehicle interface **120**. Similarly, the controller **132** (via the sub-controller **133**), or the sub-controller **133**, maintains the disconnect device **118** (assuming that the disconnect device **118** is normally open (disconnected state) and closes (connected state)) in the disconnected state which disconnects the connection between the energy storage device **122** and the towed vehicle interface **124**. In this disconnected state of the power charging system **100**, the towing vehicle interface **120** and towed vehicle interface **124** are safely unpowered. The controller **132** is programmed to only power the towing vehicle interface **120** and towed vehicle interface **124** when detecting that the towing vehicle interface **120** and towed vehicle interface **124** are connected, as described herein.

(44) Prior to connecting the towing vehicle interface **120** and towed vehicle interface **124**, the controller **132** and sub-controller **133** may actuate the safe state indicator **148**, such as the light **160** on the towing vehicle interface **120**, the light **162** on the towed vehicle interface **124**, the light on the dashboard of the towed vehicle, a notification on the display device **152**, and/or a notification

on a software app on a portable computing device **154**, that towing vehicle interface **120** and towed vehicle interface **124** are safely unpowered, and that it is safe to connect them, for example by using the connection cable **126**.

(45) The operator then connects the towing vehicle interface **120** and towed vehicle interface **124**, either directly, or using the connection cable **126**. The controller **132** detects that the towing vehicle interface **120** and towed vehicle interface **124** have been connected using from signals from the signal wire **144***v* (i.e., the controller detecting that the controller **132** is now grounded via the signal wire **144***v*), the towing vehicle disconnect sensor **176**, and/or the towing vehicle disconnect sensor **176**. Upon detecting that the towing vehicle interface **120** and towed vehicle interface **124** have been connected, the controller **132** can actuate the physical safety interlock **168** and the physical safety interlock **170** thereby locking the connection and preventing disconnection of the towing vehicle interface **120** from the towed vehicle interface **124** via the connection cable **126**, if utilized, until the power charging system **100** is in a safe state for disconnection when both the towing vehicle interface **120** from the towed vehicle interface **124** have been de-powered. Alternatively, the controller **132** can actuate the physical safety interlock **168** and the physical safety interlock **170** only after the towing vehicle interface **120** and the towed vehicle interface **124** have been connected and powered with charging power, as described herein. Again, the controller locks the physical safety interlock **168** and the physical safety interlock **170**, thereby locking the connection and prevents disconnection until the towing vehicle interface **120** from the towed vehicle interface **124** have been de-powered.

(46) The controller **132** also receives operational data from the towing vehicle sensors **136**, the towing vehicle computer **138**, the charging device **112**, the operator interfaces **146**, the towed vehicle sensors **137**, the SDMS **138**, the energy storage device **122**, and/or the sub-controller **133**. The controller **132** then determines whether to provide charging power from the charging device **112** to the energy storage device **122** based on the operational data. For example, the controller **132** may determine to provide charging power from the charging device **112**, under one or more of the following conditions based on the operational data:

(47) 1. The energy storage device **122** is not fully charged, and requires charging. This may be determined by the controller **132** based on receiving a “charge enabled” signal from the storage device charge requested signal device **137***e* via the signal wire **144***t*, and/or based on the contents of a received encoded battery state signal from the encoded storage device state signal device **137***f* via the signal wire **144** *u*, and/or based on receiving operational data regarding the operational status of the charging energy storage device from the SDMS **138** via the signal wire **144***m*, sub-controller **133** via the signal wire **142**, and/or other components of the power charging system **100**.

(48) 2. The energy storage device **122** is in condition for being charged. For example, the controller **132** may determine whether the energy storage device is at a proper temperature, non-fault condition, and/or other state for being safely charged. This may be determined by the controller **132** based on receiving operational data from: the towed vehicle sensors via the signal wires **144**, the SDMS **138** via the signal wire **144***m*, the sub-controller **133** via the signal wire **142**, and/or other components of the power charging system **100**.

(49) 3. The charging device **112** is in condition to provide charging power to the energy storage device **122**. For instance, the controller **132** may determine whether the vehicle motor is running and in a state (e.g., transmission in “drive,” “park,” or “neutral”) so that it may power the charging device (e.g., an alternator powered by the vehicle motor), whether the charging device is in a fault state, etc. This may be determined by the controller based on receiving operational data from: the charging device **112** via signal wire **144***d*, the regulator via the signal wire **144***c*, the towing vehicle sensors **136** via the signal wires **144**, and/or the towing vehicle computer **138** via signal wire **144***e*.

(50) When the controller **132** determines that it is appropriate to charge the energy storage device **122**, the controller **132** actuates the disconnect device **118** to connect the charging device **112** to the towing vehicle interface **120** and actuates the disconnect device **134** to connect the energy storage

device **122** to the towed vehicle interface **124**, thereby completing the closed "charging circuit" between the charging device **112** and the energy storage device **122**. The controller **132** then activates the charging device **112** to transfer charging power from the charging device **112** to the energy storage device **122**. The controller **132** can also control the charging device **112** and/or converter **114** to set the charging power output (e.g., the voltage and current output) by the charging device **112** and/or converter **114** to properly charge the energy storage device **122** based upon the operational data received by the controller **132**. For instance, the controller **132** may control the regulator **116** to control the power output of the charging device **112** (e.g., an alternator) and/or the converter **114**. In the case of self-actuating physical safety interlocks **168** and **170**, upon powering the charging circuit, the physical safety interlock **168** and the physical safety interlock **170** automatically lock thereby locking the connection and preventing disconnection of the towing vehicle interface **120** from the towed vehicle interface **124** via the connection cable **126**, if utilized, until the towing vehicle interface **120** from the towed vehicle interface **124** are de-powered. In addition, the controller **132** sets the safe state indicator(s) to indicate it is not safe to disconnect the towing vehicle interface **120**, the towed vehicle interface **124** and/or the connection cable **126**. For example, in the case of indicator lights **160** and **162**, the controller **132** can turn out the lights when the charging circuit is powered to indicate it is not safe to disconnect, and turn on the lights when the charging circuit is de-powered to indicate it is safe to disconnect, or vice versa.

(51) As the power charging system **100** charges the energy storage device **122**, the controller **132** continues to monitor the operational data of the power charging system **100**. For example, the controller **132** monitors the operational data to determine whether the energy storage device **122** is fully charged, as does the SDMS **138**. If the controller **132** and/or SDMS **138** determines that the storage device **122** is fully charged, the controller **132** can discontinue transferring the charging power from the charging device **112** to the energy storage device **122**. To discontinue the charging, the controller **132** may first gradually de-power the charging device **112**, such as by reducing the power output of the charging device **112**. This can protect the charging device **112** by avoiding rapid removal of the load on the charging device **112** which may damage the charging device **112** (e.g., damage to an alternator caused by rapidly removing the load). Next, the controller **132** may actuate the disconnect device **118** to disconnect the charging device **112** from the towing vehicle interface **120** and actuate the disconnect device **134** to disconnect the energy storage device **122** from the towed vehicle interface **124**. This may be done as a safety precaution in case an operator decides to manually disconnect the towing vehicle interface **120** from the towed vehicle interface **124**.

(52) The controller **132** continues to monitor the operational data to determine whether to again provide charging power from the charging device **112** to the energy storage device **122** based on the operational data, in the same manner described above. For example, as the energy storage device **122** is used to power equipment on the towed vehicle **104**, the energy storage device **122** may again require charging. When the controller **132** determines to again provide charging power from the charging device **112** to the energy storage device **122** based on the operational data, the controller performs the same charging process described above.

(53) In order to disconnect the power charging system **100**, such as when disconnecting the towed vehicle **104** from the towing vehicle **102**, the controller **132** is configured to first de-power the charging circuit in order to avoid the danger having unsecured and open powered connectors when manually disconnecting the towed vehicle interface **120** from the towing vehicle interface **124** (e.g., by disconnecting the connection cable **126**). The charging device **112** may also be disabled or shut down to avoid damaging the charging device **112** by rapidly removing the load on the charging device **112**. The power charging system **100** may be configured in several ways to accomplish the safe shutdown and disconnection, in which the power charging system **100** may be configured to utilize any one or more of the shutdown/disconnect processes.

(54) In one simple way, the power charging system **100** may utilize a manually initiated shutdown

and disconnect procedure. The controller **132** is configured to receive a manually actuatable disconnect/shutdown input from an operator via any one of the operator interfaces **146**, or even a separate user input device. Upon receiving the disconnect/shutdown input, the controller **132** may first power down the charging device **112** to prevent damage to the charging device **112**. Next, the controller **132** actuates the disconnect device **118** to disconnect the charging device **112** from the towing vehicle interface **120** and actuates the disconnect device **134** to disconnect the energy storage device **122** from the towed vehicle interface **124**, thereby safely de-powering the towing vehicle interface **120** and the towed vehicle interface **124**. The controller **132** then de-actuates (i.e., unlocks) the physical safety interlocks **168** and **170**, thereby unlocking the connection between the towing vehicle interface **120** and the towed vehicle interface **124** (including, if utilized, connection cable **126** from the towing vehicle interface **120** and the towed vehicle interface **124**). If self-actuating, the physical safety interlocks **168** and **170** unlock upon de-powering the charging circuit. The controller **132** also actuates the safe state indicator(s) **148** showing an operator that it is now safe to disconnect the connection cable **126** from the towing vehicle interface **120** and the towed vehicle interface **124**. The towed vehicle **104** may now be disconnected from the towing vehicle **102** by disconnecting the trailer coupler **110** from the trailer hitch **108**.
(55) In another manual disconnect/shutdown procedure, an operator may begin disconnecting the towed vehicle interface **120** from the towing vehicle interface **124**. The controller **132** will detect that the towing vehicle disconnect sensor **176**, and/or the towed vehicle disconnect sensor **178** are being disconnected prior to the high-power pins/contacts **172** of the towing vehicle interface **120** disconnecting from the mating high-power pins/contacts of the towing vehicle connector **128**. The controller **132** then executes the same disconnect/shutdown procedure and the operator can disconnect the towed vehicle, as described above.
(56) Another disconnect/shutdown procedure is more automated in that the controller **132** determines that the operator may be about to disconnect the towed vehicle interface **120** from the towing vehicle interface **124**, without a manual input from the operator. In this case, the controller is configured to execute the disconnect/shutdown procedure based upon an operational characteristic of the towing vehicle **102**. The controller **132** is receiving operational data from the towing vehicle sensors **136**, the towing vehicle computer **138**, the charging device **112** and the regulator **116**. This operational data includes data representative of an operational characteristic of the towing vehicle **102**. Accordingly, the controller **132** is configured and/or programmed to execute the disconnect/shutdown procedure describe above based upon an operational characteristic of the towing vehicle. For example, the operational characteristic of the towing vehicle may be that the towing vehicle is in park, or that the towing vehicle is in neutral, based upon operational data received from the transmission position sensor **136***a* via signal wire **144***d*. In addition, the operational characteristic of the towing vehicle may be that the towing vehicle parking break has been enabled, based upon operational data received from the parking break position sensor **136***b* via signal wire **144***g*. Also, the operational characteristic of the towing vehicle may be that the towing vehicle engine is turned off or is below a maximum speed, based upon operational data received from the engine RPM sensor **136***d* or **136***e*. The operational data representative of the operational characteristic may be received by the controller **132** from the towing vehicle computer **138** or by accessing a towing vehicle internal digital communications network.
(57) Accordingly, an innovative and improved power charging system **100** has been disclosed. The power charging system **100** is capable of safely transferring high power levels, above 1 kW to 7 kW, from a charging device **112** on the towing vehicle **102** to an energy storage device **122** on the towed vehicle **104**. The power charging system **100** is also configured to safely de-power components of the system to prevent hazardous conditions which can create a risk of accidental electrical shock and/or damage to the system components.
(58) Although particular embodiments have been shown and described, it is to be understood that the above description is not intended to limit the scope of these embodiments. While embodiments

and variations of the many aspects of the invention have been disclosed and described herein, such disclosure is provided for purposes of explanation and illustration only. Thus, various changes and modifications may be made without departing from the scope of the claims. For example, not all of the components described in the embodiments are necessary, and the invention may include any suitable combinations of the described components, and the general shapes and relative sizes of the components of the invention may be modified. Accordingly, embodiments are intended to exemplify alternatives, modifications, and equivalents that may fall within the scope of the claims. The invention, therefore, should not be limited, except to the following claims, and their equivalents.

Claims

1. A power charging system for a towing vehicle and a towed vehicle connectable to the towing vehicle, the power charging system comprising: a charging device disposed on the towing vehicle, wherein the charging device includes an alternator; a towing vehicle interface disposed on the towing vehicle and in selectable electrical connection with the charging device; an energy storage device disposed on the towed vehicle; a towed vehicle interface disposed on the towed vehicle and in selectable electrical connection with the energy storage device, the towed vehicle interface configured to be electrically connected and disconnected with the towed vehicle interface; and a communication system configured to communicate operational data representative of operational conditions of the charging device and the energy storage device to a controller; wherein the controller is operably coupled to the charging device and configured to control operation of the alternator based upon the operational data; wherein the alternator of the charging device is configured to transfer charging power from the alternator to the energy storage device, and wherein the controller is configured to selectably de-power the towing vehicle interface based upon an operational characteristic of the towing vehicle, wherein the operational characteristic is one or more selected from the group of: the towing vehicle being in park; the towing vehicle being in neutral; the towing vehicle being at idle; a towing vehicle parking brake being enabled; a towing vehicle engine being turned off; and the towing vehicle being below a maximum speed.
2. The power charging system of claim 1, wherein the charging device further includes an alternator controller.
3. The power charging system of claim 1, wherein the alternator is a chassis alternator of the towing vehicle.
4. The power charging system of claim 1, wherein the alternator is dedicated to the power charging system and is not configured for powering or charging any towing vehicle equipment.
5. The power charging system of claim 1, wherein the charging device comprises a power conversion device.
6. The power charging system of claim 5, wherein the power conversion device is selected from a group consisting of a DC-DC converter, a DC-AC converter, an AC-DC converter, a pulse width modulation controller, a current limiting wire, and a current limiting self-resetting device.
7. The power charging system of claim 1, wherein the towed vehicle interface and the towing vehicle interface each comprises a single housing carrying both the charging power and operational data.
8. The power charging system of claim 7, wherein the towed vehicle interface and the towing vehicle interface each comprise high-power pins for carrying the charging power and separate auxiliary pins for carrying the operational data.
9. The power charging system of claim 1, wherein the towed vehicle interface and the towing vehicle interface each comprises a power housing for carrying the charging power, and a physically separate data housing for carrying at least some of the operational data and not carrying any of the charging power.

10. The power charging system of claim 9, wherein at least some of the operational data is not carried by the physically separate data housing.
11. The power charging system of claim 1, wherein the communication system is configured to communicate operational data using a digital communication protocol.
12. The power charging system of claim 11, wherein the communication system comprises a controller area network (CAN) having a CAN bus.
13. The power charging system of claim 11, wherein the communication system comprises one of an Ethernet network and a RS-485 network.
14. The power charging system of claim 1, wherein the communication system comprises a wireless communication protocol.
15. The power charging system of claim 14, wherein the wireless communication protocol is selected from a group consisting of WiFi, Bluetooth, wireless USB, Zigbee, and cellular phone protocol.
16. The power charging system of claim 1, wherein the communication system comprises one or more signal wires operably coupled to the controller.
17. The power charging system of claim 16, wherein the one or more signal wires comprise one or more analog sensing wires.
18. The power charging system of claim 17, wherein the one or more analog sensing wires include one or more of a battery voltage sensing wire, a battery temperature wire, a thermocouple wire, a battery current sensing wire, an energy storage device charge requested signal wire, and an encoded battery state signal wire.
19. The power charging system of claim 16, wherein the one or more signal wires are each selected from a group consisting of: a digital signal wire; a signal level wire; and an analog sensing wire.
20. The power charging system of claim 16, wherein the one or more signal wires comprise: a connection present signal wire which indicates whether the towing vehicle interface and towed vehicle interface are connected; an energy storage device charging status signal wire which indicates whether charging of the energy storage device is enabled, charging of the energy storage device is disabled, and charge state to be maintained in which the charging device supplies power to other loads in the towed vehicle and only provides power to the energy storage device to maintain a current state of charge.
21. The power charging system of claim 1, wherein the operational data includes a charging status of the energy storage device, including one of a charging enable status and a charging disabled status.
22. The power charging system of claim 21, wherein the energy storage device comprises a storage device management system (SDMS) and the charging status is determined by the SDMS and communicated from the SDMS to the controller.
23. The power charging system of claim 21, wherein the energy storage device comprises a storage device management system (SDMS) configured to communicate operational data regarding the energy storage device to the controller, and the charging status is determined by the controller based upon operational data obtained from the SDMS.
24. The power charging system of claim 21, wherein the charging status is determined by the controller based upon the operational data.
25. The power charging system of claim 21, wherein the charging status further includes charging needs of the energy storage device.
26. The power charging system of claim 25, wherein the operational data includes one or more present properties of the energy storage device, including at least one selected from a group of present state of temperature, state of charge, voltage, and amperage into or out of energy storage device.
27. The power charging system of claim 26, wherein the operational data further includes specifications of the energy storage device, present needs of the charging device and energy storage

device, and warnings, alarms and faults of both the charging device and energy storage device.

28. The power charging system of claim 1, wherein the operational data includes an operational status of the charging device, specifications of the charging device, an operational status of the energy storage device, and specifications of the energy storage device, wherein: the operational status of the charging device includes one or more properties of the charging device selected from a group of: current output; voltage output; temperature; overheating status; percentage utilization; operating mode, including standby, active, initializing, faulted state; identification of faults, including overheated, internal fault condition-logic error, energy storage device not communicating, lost communication, broken sensor, energy storage device fault; target energy storage voltage, target energy storage current; the operational status of the energy storage device includes one or more properties of the energy storage device selected from a group of: present state of charge, present energy storage capacity; present voltage; present temperature; present current; charging mode, including charging enabled or charging disable; target energy storage voltage; target energy storage current; number of battery management systems (BMS's) present; whether the energy storage device is online; fault state; and state of health; the specifications of the energy storage device include one or more selected from a group of: type of energy storage device; overall storage capacity of the energy storage device; charging current limits of the energy storage device; charging current limits of the BMS; charging voltage limits; charging temperature limits; and charging rate limits; and the specifications of the charging device include one or more of selected from a group of: output amperage capacity; output voltage; allowable operating temperature range; and power output capacity.

29. The power charging system of claim 1, wherein the controller is configured to selectably de-power the towing vehicle interface and the towed vehicle interface prior to disconnecting the towing vehicle interface from the towed vehicle interface.

30. The power charging system of claim 29, wherein the controller is configured to only power the towing vehicle interface and the towed vehicle interface when the towing vehicle interface is connected to the towed vehicle interface.

31. The power charging system of claim 29, wherein the controller is configured to selectably de-power the towing vehicle interface and the towed vehicle interface based upon a manually actuatable input by an operator.

32. The power charging system of claim 31, further comprising a safe state indicator which indicates when it is safe to disconnect the towing vehicle interface and the towed vehicle interface upon de-powering of the towing vehicle interface and the towed vehicle interface.

33. The power charging system of claim 32, wherein the safe state indicator is one of a light on one of the towing vehicle interface and/or the towed vehicle interface, a light on a dashboard of the towed vehicle, a notification on a display device, and a notification on a software app on a handheld computing device.

34. The power charging system of claim 29, wherein the controller is further configured to selectably de-power the towed vehicle interface based upon the operational characteristic of the towing vehicle.

35. The power charging system of claim 1, wherein the operational characteristic is determined by one or more of: accessing a towing vehicle internal digital communications network; use of switches indicating position of transmission of the towing vehicle; use of switches indicating position of parking brake mechanism of the towing vehicle; use of sensors to sense motion of the towing vehicle; use of sensors coupled to the charging device to indicate engine RPMs of the towing vehicle; and use of sensors to indicate RPM of engine or transmission of the towing vehicle.

36. The power charging system of claim 35, further comprising a safe state indicator which indicates when it is safe to disconnect the towing vehicle interface and the towed vehicle interface upon de-powering of the towing vehicle interface and the towed vehicle interface.

37. The power charging system of claim 36, wherein the safe state indicator is one of a light on one

of towing vehicle interface and/or the towed vehicle interface, a light on a dashboard of the towed vehicle, a notification on a display device, and a notification on a software app on a handheld computing device.

38. The power charging system of claim 1, wherein at least one of the towing vehicle interface and the towed vehicle interface comprises a physical safety interlock which prevents disconnection of the towing vehicle interface and the towed vehicle interface until the power charging system and energy storage device are in a safe state for disconnection.

39. The power charging system of claim 38, wherein the physical safety interlock is operably coupled to, and controlled by the controller.

40. The power charging system of claim 1, wherein: at least one of the towing vehicle interface and the towed vehicle interface comprises a disconnect sensor configured to detect that the towing vehicle interface and the towed vehicle are being disconnected; the disconnect sensor operably coupled to the controller; and the controller configured to de-power the towing vehicle interface and the towed vehicle interface upon detecting that the towing vehicle interface and the towed vehicle are being disconnected.

41. The power charging system of claim 40, wherein the towing vehicle interface, the towed vehicle interface and the disconnect sensor are configured such that the disconnect sensor detects that towing vehicle interface and the towed vehicle interface are being disconnected prior to a power connection between the towing vehicle interface and the towed vehicle interface carrying charging power from the charging device is disconnected.

42. The power charging system of claim 39, wherein the controller is configured to de-power the towing vehicle interface and the towed vehicle interface based upon a disconnect sensor detecting that the towing vehicle interface and the towed vehicle interface are being disconnected prior to de-powering the towing vehicle interface and the towed vehicle interface.

43. The power charging system of claim 1, further comprising a safe state indicator which indicates when it is safe to disconnect the towing vehicle interface and the towed vehicle interface upon de-powering of the towing vehicle interface and the towed vehicle interface.

44. The power charging system of claim 43, wherein the safe state indicator is one of a light on one of towing vehicle interface and/or the towed vehicle interface, a light on a dashboard of the towed vehicle, a notification on a display device, and a notification on a software app on a handheld computing device.

45. The power charging system of claim 1, further comprising a human user interface configured to communicate a status of the power charging system.

46. The power charging system of claim 45, wherein the human user interface is integrated with a fleet vehicle monitoring system.

47. The power charging system of claim 45, wherein the human user interface comprises one of: a dashboard display on the towing vehicle; one or more indicator lights; a graphical display device; an LCD display; an LED display; an OLED display; and a wireless communication module configured to communicate with a software application (app) on a handheld computing device.

48. The power charging system of claim 47, wherein the human user interface is integrated with a fleet vehicle monitoring system.

49. The power charging system of claim 48, wherein the human user interface is dedicated to the power charging system.

50. The power charging system of claim 1, wherein the power charging system is configured to transfer charging power of at least 1 kW from the charging device to the energy storage device.

51. The power charging system of claim 1, wherein the power charging system is configured to transfer charging power of at least 4 kW from the charging device to the energy storage device.

52. The power charging system of claim 1, wherein the power charging system is configured to transfer charging power of at least 5 kW from the charging device to the energy storage device.

53. The power charging system of claim 1, wherein the power charging system is configured to

transfer charging power of at least 7 kW from the charging device to the energy storage device.
54. A method for charging an energy storage device on a towed vehicle, the method comprising: obtaining operational data representative of operational conditions of an alternator of a charging device on a towing vehicle and an energy storage device on a towed vehicle; controlling operation of the alternator based at least in part on the operational data; transferring charging power from the alternator to the energy storage device; and selectably de-powering the alternator based upon an operational characteristic of the towing vehicle, wherein the operational characteristic is one or more selected from the group of: the towing vehicle being in park; the towing vehicle being in neutral; the towing vehicle being at idle; a towing vehicle parking brake being enabled; a towing vehicle engine being turned off; and the towing vehicle being below a maximum speed.
55. The method of claim 54, wherein transferring charging power from the alternator to the energy storage device includes transferring charging power of at least 1 kW from the alternator to the energy storage device.
56. The method of claim 54, wherein transferring charging power from the alternator to the energy storage device does not include powering or charging any equipment of the towing vehicle.
57. The method of claim 54, further comprising converting power of the charging device using a power conversion device.
58. The method of claim 57, wherein the power conversion device is selected from a group consisting of a DC-DC converter, a DC-AC converter, an AC-DC converter, a pulse width modulation controller, a current limiting wire, and a current limiting self-resetting device.
59. The method of claim 54, further comprising communicating the obtained operational data using auxiliary pins which are separate from high power pins for carrying the transferred charging power.
60. The method of claim 54, comprising communicating the obtained operational data using a wireless communication protocol.
61. The method of claim 60, wherein the wireless communication protocol is selected from a group consisting of WiFi, Bluetooth, wireless USB, Zigbee, and cellular phone protocol.
62. The method of claim 54, further comprising providing an indication that the charging device and the energy storage device are electrically connected.
63. The method of claim 54, wherein the operational data includes a charging status of the energy storage device, including one of a charging enable status and a charging disabled status.
64. The method of claim 54, wherein the operational data includes one or more present properties of the energy storage device, including at least one selected from a group of present state of temperature, state of charge, voltage, and amperage into or out of the energy storage device.
65. The method of claim 54, further comprising determining the operational characteristic by one or more of: accessing a towing vehicle internal digital communications network; indicating a position of a transmission of the towing vehicle using switches; indicating a position of a parking brake mechanism of the towing vehicle using switches; sensing motion of the towing vehicle; sensing engine RPMs of the towing vehicle; and sensing RPM of an engine or transmission of the towing vehicle.